

Weighted-RVB Initialization and Heisenberg-HVA Refinement for a 19-Site Kagome Antiferromagnetic Heisenberg Benchmark

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Abstract

We present a finite-size proof-of-concept for preparing low-energy states of a 19-site Kagome antiferromagnetic Heisenberg model using a signed weighted resonating-valence-bond (RVB) initializer followed by shallow edge-colored Heisenberg-HVA layers. The model is benchmarked by exact sparse diagonalization in the fixed- S^z sector with $n_{\downarrow} = 9$ and Hilbert-space dimension 92,378.

A static 9-singlet product state gives energy $E = -27.000000$ and fidelity $F = 0.014241$ against the exact ground state. Optimizing signed amplitudes over 54 maximum dimer coverings improves this to $E = -29.012072100265$ and $F = 0.978535993636$. A no-calibration, edge-colored Heisenberg-HVA refinement reaches $E = -29.037601351802$ and $F = 0.982518440088$ at depth $p = 4$, closing 94.94% of the static-dimer-to-exact energy gap. Exact calibrated-Hamiltonian reference scans can reach closer finite-size states, with the best reproduced triangle perturbation giving error 3.70×10^{-5} and fidelity 0.999966. The result should therefore be interpreted as a strong no-calibration circuit-refinement baseline and not as a scalable or hardware-ready proof of quantum spin liquid preparation.

1 Introduction

The spin-1/2 Kagome antiferromagnetic Heisenberg model is a canonical frustrated-magnetism benchmark and a central model in the search for quantum spin liquid (QSL) behavior. Large-scale DMRG studies support a gapped topological spin-liquid ground state and place the thermodynamic-limit energy near $-0.4386(5)$ per site in the physical spin convention [1, 2]. The open-boundary 19-site patch studied here is small enough for exact fixed-sector diagonalization yet large enough to expose limitations of purely local dimer product states.

This report studies the following workflow:

weighted RVB initialization $\longrightarrow$ edge-colored Heisenberg-HVA refinement
 $\longrightarrow$ 19-site exact benchmark comparison.

The central question is whether improving the variational state structure can approach the exact 19-site ground state without relying on Hamiltonian calibration. The answer in this proof-of-concept is cautiously positive: the no-calibration HVA improves energy, fidelity, magnetization, entropy, and bond-correlation diagnostics, but it remains above the best exact calibrated-Hamiltonian reference states.

Figure 1 gives the project flow as two connected branches. The exact-reference branch diagonalizes the fixed 19-site Kagome Hamiltonian to obtain E_0 and $|\psi_0\rangle$. The ansatz branch constructs a signed weighted-RVB initializer, refines it with edge-colored Heisenberg-HVA layers, and compares the resulting trial state with the exact reference through energy, fidelity, and correlation diagnostics. The dashed calibration-scan box is included only as design feedback for future local HVA blocks.

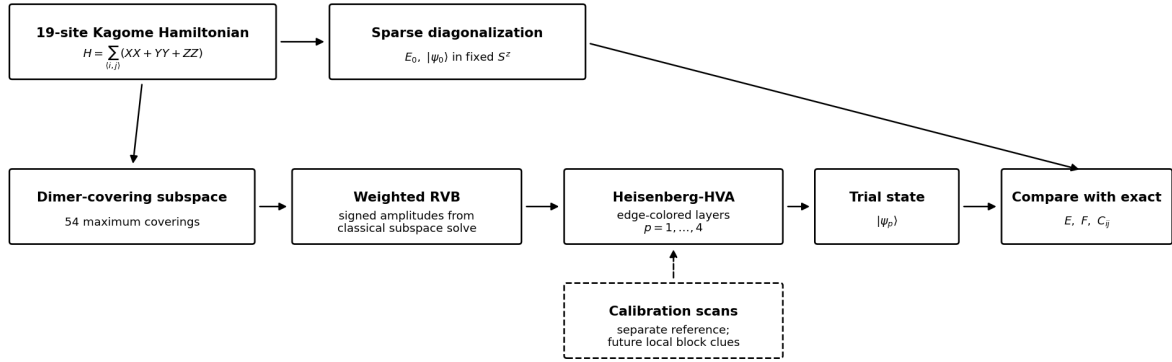


Figure 1: Project schematic. The target Hamiltonian is fixed throughout. The upper branch produces the exact reference state, and the lower branch produces the no-calibration variational trial state through weighted-RVB initialization and Heisenberg-HVA refinement. Calibration scans are shown with a dashed box because they are future ansatz-design guidance, not a replacement for the present no-calibration workflow.

2 Provenance and Scope

The original Ahsan Kagome-VQE material is used as provenance and methodology context, not as code copied into this workflow. In particular, it provides the source cross-check for the 19-site Kagome connectivity, context for hardware-efficient Qiskit VQE experiments, and the calibrated-Hamiltonian or defect-triangle strategy described in Ref. [7]. The present proof-of-concept instead develops a separate no-calibration route based on signed weighted-RVB initialization and shallow edge-colored Heisenberg-HVA refinement.

Table 1: Role of each component in the final project narrative.

Component	Role
External Ahsan Kagome-VQE material	Original calibrated-Hamiltonian/VQE provenance and future hardware-method context.
19-site bond list	Confirmed against the Ahsan notebook and used as the active benchmark graph.
Weighted RVB	Main physics initializer and dominant improvement over static dimers.
Heisenberg-HVA	No-calibration circuit-refinement path.
Calibration scans	Exact-state reference comparison, not the main no-calibration claim.
ODR / IBM Runtime ideas	Future hardware and noise-mitigation extension.

When discussing the original calibrated-Hamiltonian direction, we use cautious language: defect or triangle couplings are increased above the uniform value, with reported calibrated values depending on the patch, Hamiltonian convention, defect set, and scan. In the Ahsan workflow this idea appears as defect-triangle Hamiltonian engineering with selected bonds near $J' \approx 2$ and a hardware-efficient RealAmplitudes-style VQE circuit [7]. This report states precise calibration values only for scans reproduced in the current repository.

3 Model and Benchmark

The active Hamiltonian uses the unscaled Pauli convention

$$H = \sum_{\langle i,j \rangle \in E} (X_i X_j + Y_i Y_j + Z_i Z_j), \quad (1)$$

where E is the 30-bond 19-site Kagome patch. Physical spin energies are one quarter of Eq. (1):

$$H_{\text{phys}} = \sum_{\langle i,j \rangle} \mathbf{S}_i \cdot \mathbf{S}_j = \frac{1}{4} H.$$

The exact reference is the lowest eigenstate in the $n_{\downarrow} = 9$ fixed- S^z sector:

$$E_0 = -29.146168109350.$$

The sector dimension is

$$\dim \mathcal{H}_{n_{\downarrow}=9} = \binom{19}{9} = 92,378.$$

In the rounded convention used in the Ahsan calibration table, the corresponding 19-site value appears as approximately -29.14 [7]. All reported fidelities use

$$F(\psi) = |\langle \psi_{\text{exact}} | \psi \rangle|^2. \quad (2)$$

3.1 Edge Coloring

The 30 bonds are partitioned into four disjoint matchings. Within each matching, all two-qubit Heisenberg gates commute on disjoint qubits and may be applied in parallel. This converts a bond-dependent layer with 30 parameters into an edge-colored HVA layer with only four parameters.

Table 2: Four-color edge grouping used by the HVA. Bonds inside each group are disjoint.

Group	Bonds
0	(1, 2), (4, 6), (8, 10), (11, 12), (14, 16), (17, 18)
1	(0, 1), (2, 4), (6, 8), (10, 11), (12, 14), (16, 17)
2	(2, 3), (4, 5), (6, 7), (8, 9), (10, 12), (13, 14), (15, 16), (1, 11), (0, 17)
3	(3, 4), (5, 6), (7, 8), (9, 10), (12, 13), (14, 15), (16, 18), (2, 11), (1, 17)

4 Variational States

4.1 Static Dimer Baseline

The simplest control state is a product of nine singlets over a greedy maximum matching, leaving one unpaired spin in the 19-site odd patch:

$$|s_{ij}\rangle = \frac{|0_i 1_j\rangle - |1_i 0_j\rangle}{\sqrt{2}}.$$

This state is hardware transparent but too rigid: it reaches $E = -27.000000$ and has fidelity only $F = 0.014241$.

4.2 Signed Weighted-RVB Initializer

The RVB ansatz is a linear combination of all 54 deterministic maximum dimer coverings:

$$|\psi_{\text{RVB}}(c)\rangle = \sum_{\alpha=1}^{54} c_{\alpha} |D_{\alpha}\rangle. \quad (3)$$

The coefficients are obtained classically by solving the generalized eigenproblem

$$H_{\alpha\beta} c_{\beta} = E S_{\alpha\beta} c_{\beta}, \quad H_{\alpha\beta} = \langle D_{\alpha} | H | D_{\beta} \rangle, \quad S_{\alpha\beta} = \langle D_{\alpha} | D_{\beta} \rangle. \quad (4)$$

The equal positive RVB superposition is a poor approximation, but optimized signs and amplitudes are essential: the weighted RVB state reaches $E = -29.012072100265$ and $F = 0.978535993636$.

4.3 Heisenberg-HVA Refinement

Starting from the weighted-RVB initializer, the edge-colored HVA applies Heisenberg interactions with one shared parameter per color and layer:

$$|\psi_p(\theta)\rangle = \left[\prod_{\ell=1}^p \prod_{c=0}^3 \prod_{(i,j) \in E_c} \exp[-i\theta_{\ell,c}(X_i X_j + Y_i Y_j + Z_i Z_j)] \right] |\psi_{\text{RVB}}\rangle. \quad (5)$$

Thus the depth- p edge-colored HVA has $4p$ parameters. The reported serious run uses a Nelder-Mead optimization with $p = 0, 1, 2, 3, 4$ cached in the current release artifacts.

Figure 2 makes the circuit explicit. One HVA layer consists of four edge-color blocks. A color block applies the same variational angle to all disjoint bonds in that color group. In Qiskit, each interaction

$$\exp[-i\theta(X_i X_j + Y_i Y_j + Z_i Z_j)]$$

is implemented as $\text{RXX}(2\theta)$, $\text{RYY}(2\theta)$, and $\text{RZZ}(2\theta)$ on the same qubit pair. The implementation is `src/kagome_heisenberg_poc.py::build_heisenberg_ansatz`, and a compact standalone circuit description is included in `docs/ansatz_circuit.md`.

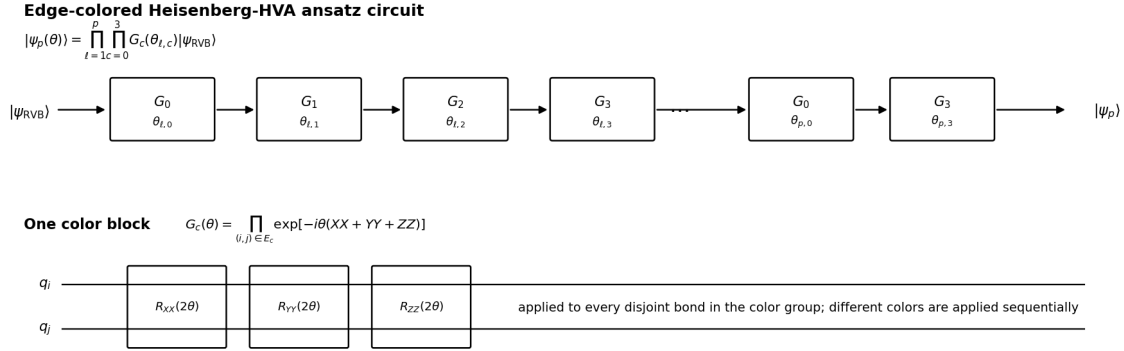


Figure 2: Edge-colored Heisenberg-HVA ansatz circuit. The weighted-RVB initializer is followed by repeated color blocks $G_c(\theta_{\ell,c})$. Each color block is a product of disjoint two-qubit Heisenberg interactions, decomposed into Qiskit Pauli rotation gates $R_{XX}(2\theta)$, $R_{YY}(2\theta)$, and $R_{ZZ}(2\theta)$.

5 Diagnostics

The analysis tracks more than energy. For each state, the repository computes fidelity, selected bipartite entropies, maximum site magnetization, bond correlations, and bond-correlation errors relative to the exact state.

For a bond (i, j) , the plotted correlation is

$$C_{ij} = \langle X_i X_j + Y_i Y_j + Z_i Z_j \rangle. \quad (6)$$

A perfect singlet has $C_{ij} = -3$. The antiferromagnetic bond participation diagnostic is

$$p_{ij} = \frac{\max(0, -C_{ij})}{\sum_{(k,l)} \max(0, -C_{kl})}, \quad P_{\text{AF}} = \frac{1}{\sum_{ij} p_{ij}^2}. \quad (7)$$

Large P_{AF} indicates more distributed antiferromagnetic correlation weight. Motivated by the DMRG diagnostics of Ref. [2], the code also reports the normalized participation $P_{\text{AF}}/|E|$, the fraction of antiferromagnetic weight trapped on nearly perfect singlet bonds with $C_{ij} < -2$, and a graph-distance spin-correlation profile. The fitted graph-distance decay length is only a finite-patch proxy; it is not a thermodynamic correlation length. The figures use a deterministic graph layout for visualization, not a physical Kagome embedding.

6 Results

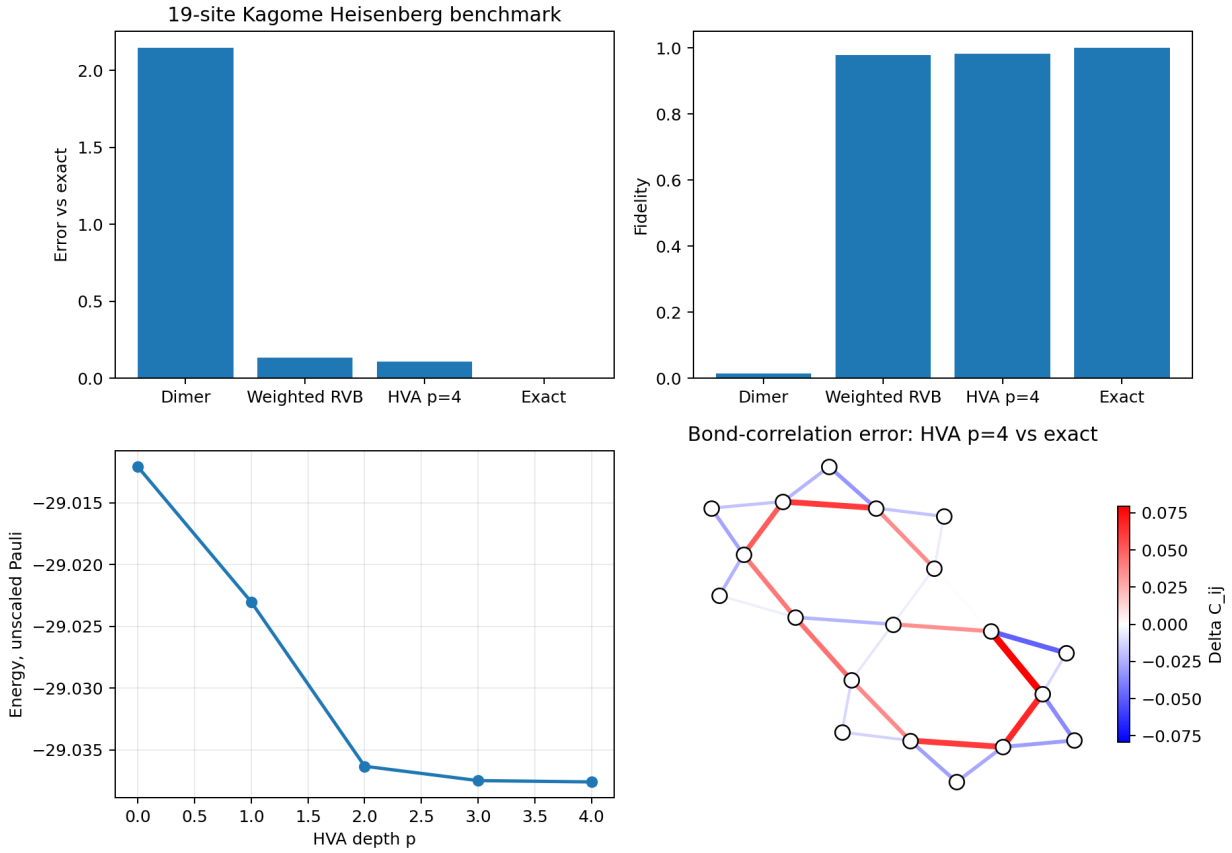


Figure 3: One-page summary of the 19-site benchmark. The weighted-RVB initializer dramatically improves on the static dimer product state, and Heisenberg-HVA refinement gives a smaller but consistent no-calibration improvement through $p = 4$.

Table 3: Main exact-benchmark results in the unscaled Pauli convention. Error is relative to $E_0 = -29.146168109350$.

State	Depth	Energy	Error	Fidelity	Gap closed	Entropy	Max mag.
Static dimer	0	-27.00000000	2.14616811	0.014241	0.00%	1.000000	0.500000
Equal RVB-54	0	-26.39781307	2.74835504	0.004371	-28.06%	3.366154	0.129849
Weighted RVB-54	0	-29.01207210	0.13409601	0.978536	93.75%	3.129502	0.107755
Weighted RVB + HVA	1	-29.02302627	0.12314184	0.979995	94.26%	3.129583	0.107528
Weighted RVB + HVA	2	-29.03633182	0.10983629	0.982446	94.88%	3.140691	0.104773
Weighted RVB + HVA	3	-29.03749397	0.10867414	0.982442	94.94%	3.141921	0.104388
Weighted RVB + HVA	4	-29.03760135	0.10856676	0.982518	94.94%	3.155489	0.104055
Exact	–	-29.14616811	0.00000000	1.000000	100.00%	3.212669	0.094007

Table 4: Paper-inspired correlation diagnostics. The frozen AF fraction measures how much antiferromagnetic bond weight lies on nearly perfect dimers. The graph-distance decay length is a compact finite-patch proxy inspired by the log-linear correlation plots of Ref. [2].

State	$P_{\text{AF}}/30$	Frozen AF frac.	ξ_{graph}
Static dimer	0.300	1.000	–
Weighted RVB-54	0.681	0.000	1.662
Weighted RVB + HVA $p = 4$	0.681	0.000	1.624
Exact	0.682	0.000	1.356

6.1 Depth Dependence

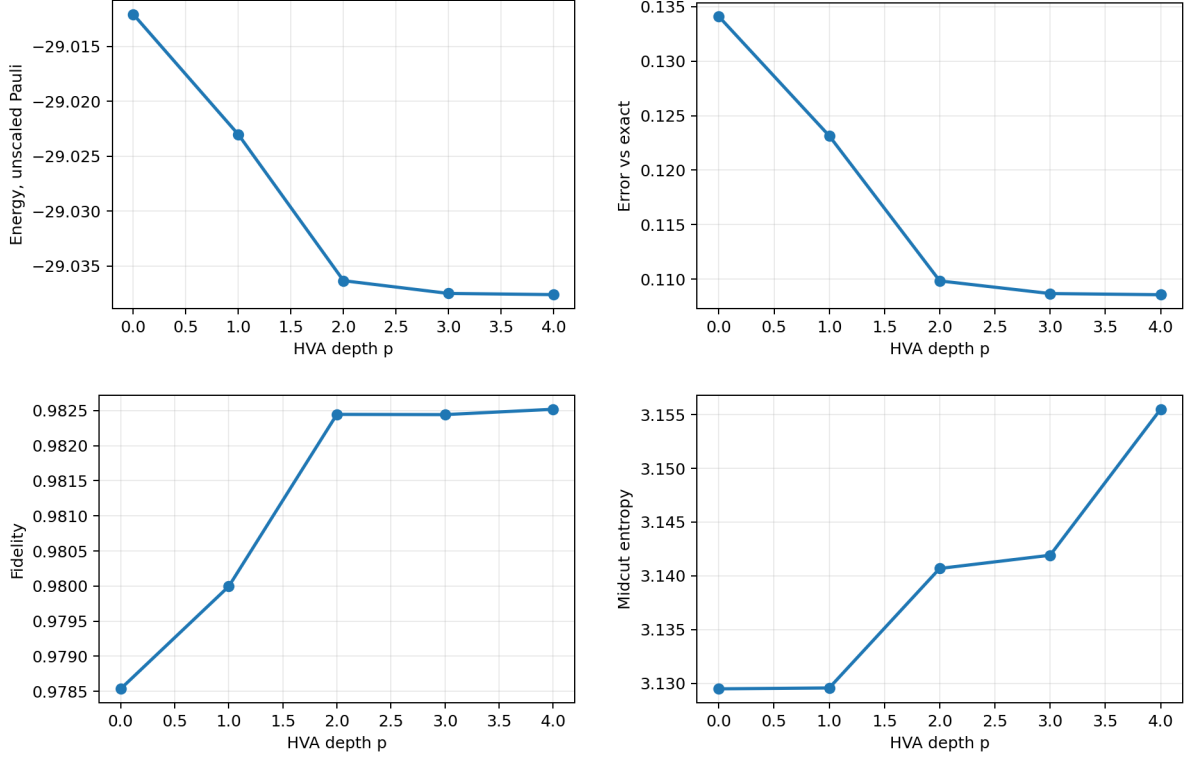


Figure 4: Depth dependence of energy, error, fidelity, and entropy. The largest improvement comes from weighted-RVB initialization; HVA layers provide a smaller additional circuit-compatible refinement.

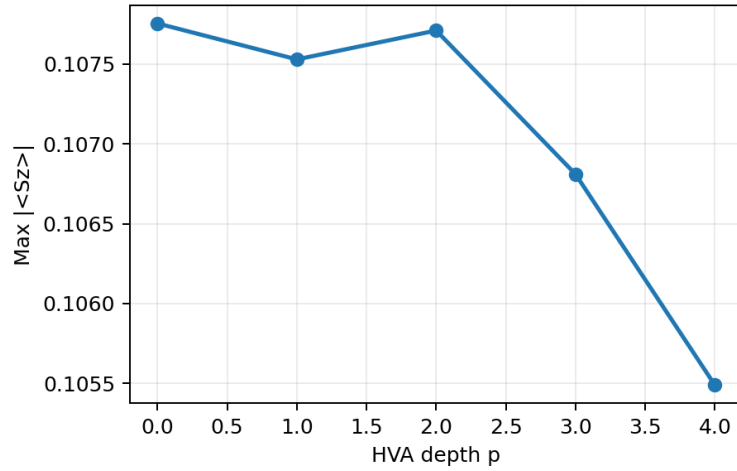


Figure 5: Maximum site magnetization versus HVA depth. The weighted RVB and HVA-refined states remain much closer to the low-magnetization exact state than the static dimer baseline.

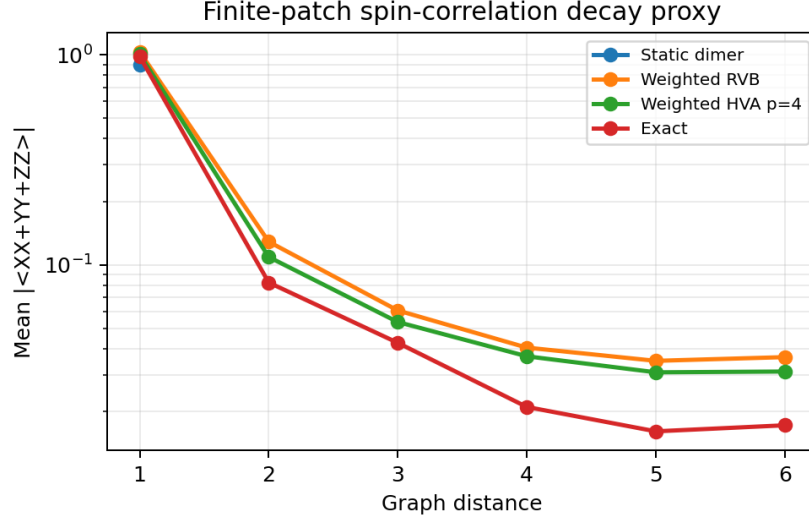


Figure 6: Finite-patch spin-correlation decay proxy. Correlations are averaged by graph distance because reliable physical coordinates for the extracted 19-site patch are not part of the input data. The static dimer has only nearest-neighbor singlet correlations, while the weighted-RVB and HVA-refined states show distributed short-range correlations closer to the exact-state pattern.

6.2 Bond-Correlation Structure

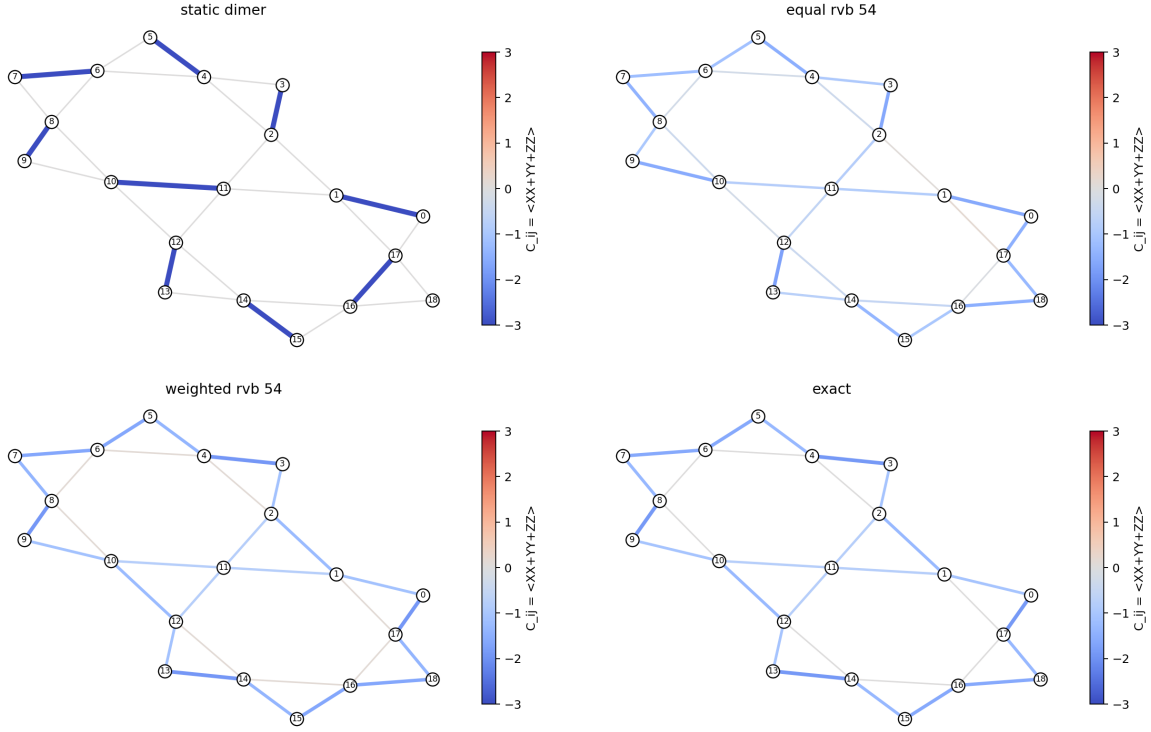


Figure 7: Bond correlations for the static dimer baseline, equal RVB, weighted RVB, and exact state. The weighted-RVB state captures the exact correlation pattern much better than either the static product state or equal positive RVB.

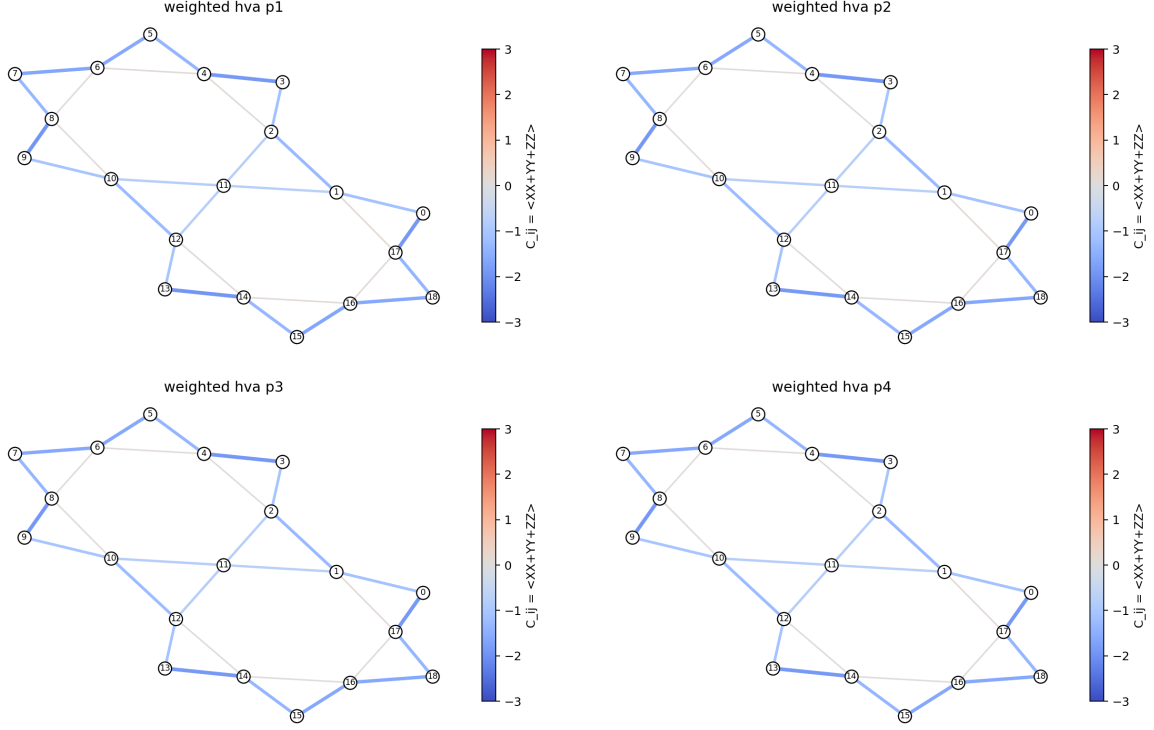


Figure 8: Bond-correlation maps after HVA refinement at $p = 1, 2, 3, 4$. Refinement is incremental but consistently preserves the distributed antiferromagnetic structure of the weighted-RVB initializer.

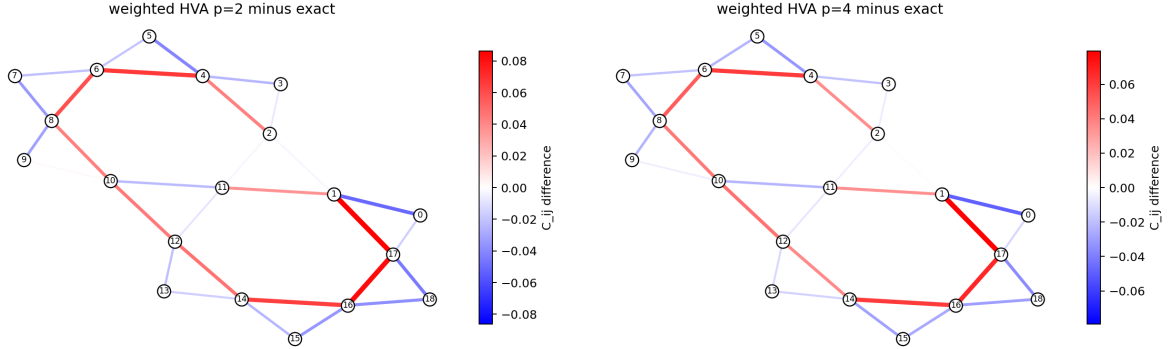


Figure 9: Bond-correlation error maps relative to the exact state. Left: earlier $p = 2$ HVA reference. Right: best current no-calibration HVA, $p = 4$.

6.3 Calibration Reference Comparison

The no-calibration HVA result should be compared honestly against exact calibrated-Hamiltonian reference scans. In those scans, the ground state of a modified Hamiltonian is obtained exactly and then evaluated against the original target Hamiltonian. This is a reference-state diagnostic, not a direct hardware-preparable circuit comparison.

Table 5: No-calibration HVA compared with reproduced calibrated-Hamiltonian exact reference states. Calibrated rows are exact ground states of modified Hamiltonians evaluated against the original target Hamiltonian.

Method / reference state	Energy error	Fidelity
Weighted RVB + HVA $p = 4$	0.108567	0.982518
Calibrated group color1, $J' = 1.02$	0.003945	0.994967
Calibrated bond 14–16, $J' = 1.02$	0.000510	0.999626
Calibrated triangle 6–7–8, $J' = 1.02$	0.000037	0.999966

Figure 10 plots the same comparison as a landscape. The horizontal axis is target energy error, $|E - E_{\text{exact}}|$, so moving left means better energy agreement with the original 19-site Hamiltonian. The vertical axis is fidelity with the exact target state, so moving up means better wavefunction agreement. The best region is therefore the upper-left corner.

The colored scan points in Fig. 10 are exact ground states of modified Hamiltonians: a bond, edge-color group, or triangle has coupling J' instead of the uniform value. The red star is the current no-calibration weighted-RVB + HVA $p = 4$ state. This distinction matters. The figure does not show that the present no-calibration circuit beats calibration. Instead, it shows that small local Hamiltonian changes, especially on triangles, can produce exact reference states extremely close to the target. Those local structures are useful clues for future HVA blocks, while the target Hamiltonian used for evaluation remains unchanged.

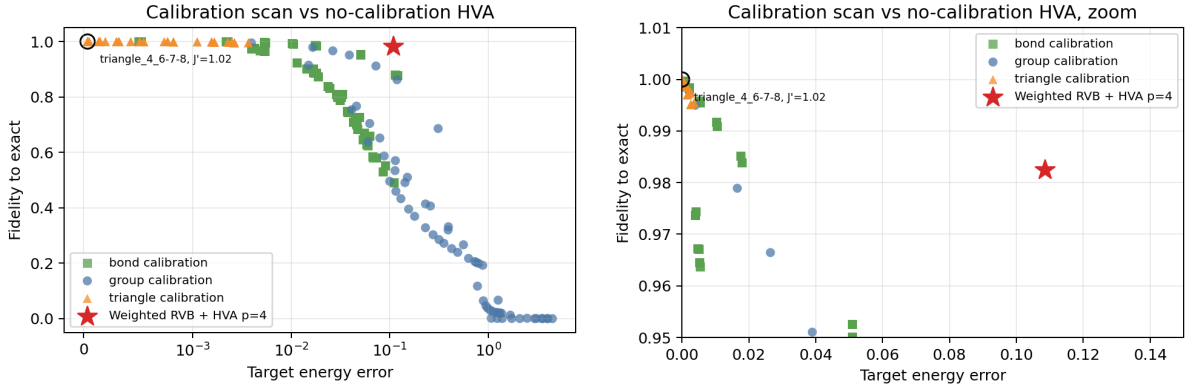


Figure 10: Calibration scan landscape. Left means lower target-energy error and up means higher exact-state fidelity, so the upper-left corner is best. The colored points are exact ground states of modified Hamiltonians; the red star is the present no-calibration weighted-RVB + HVA $p = 4$ result. The figure motivates calibration-informed ansatz design, not a claim that the current HVA already outperforms calibrated reference states.

7 Discussion

The main scientific observation is that the signs and amplitudes of dimer coverings matter strongly. Equal RVB over the 54 maximum coverings performs worse than a static dimer product, while signed weighted RVB jumps to high fidelity. This indicates that the relevant finite-size spin-liquid-

like structure is not merely a uniform superposition of local singlet coverings; it requires phase and amplitude structure in the dimer-covering manifold.

The HVA contribution is more modest but important. The $p = 4$ edge-colored Heisenberg-HVA result improves the weighted-RVB energy from -29.01207210 to -29.03760135 and fidelity from 0.978536 to 0.982518 without Hamiltonian calibration. This is not enough to match the exact calibrated-Hamiltonian references, but it demonstrates that a physically structured two-qubit Heisenberg ansatz can refine a high-quality RVB state in the direction of the exact finite-size ground state.

The result is also useful because it separates two routes that were previously entangled in the project narrative:

state-structure improvement versus Hamiltonian calibration.

The calibrated-Hamiltonian route remains powerful as an exact-state reference. The no-calibration HVA route is the cleaner circuit-refinement proof-of-concept.

8 Limitations

- The result is finite-size only and uses a 19-site open-boundary patch.
- The exact benchmark is in the fixed $S^z = 1/2$ sector.
- The weighted-RVB initializer is obtained by a classical generalized eigenproblem and is not yet a hardware-preparable state-preparation circuit.
- The Heisenberg-HVA improvement is real but modest through $p = 4$.
- The no-calibration HVA does not outperform the best exact calibrated-Hamiltonian reference scan in energy or fidelity.
- No repeated-optimizer statistical robustness campaign is included in this frozen package.
- No noise model, Qiskit Runtime execution, or hardware experiment is included.
- No scalable QSL preparation claim is made.

9 Reproducibility

The main repository artifacts are:

- `data/19site_bonds.csv`: active 19-site bond list and edge coloring.
- `results/final_result_table.csv`: main result table.
- `results/19site_paper_diagnostics.csv`: paper-inspired correlation summaries.
- `results/19site_spin_distance_profile.csv`: graph-distance spin-correlation profile used in Fig. 6.

- `results/literature_benchmark_comparison.csv`: convention conversion against Depenbrock DMRG and Ahsan rounded values.
- `results/no_calibration_vs_best_calibration.csv`: compact comparison of the best HVA and best calibration reference.
- `results/19site_heisenberg_hva_fast_weighted_p4_neldermead*.csv`: current best serious no-calibration HVA output.
- `figures/`: all plots included in this report.

The default quick reproduction path is:

```
.\run_all.ps1
```

or, on Unix-like systems,

```
bash run_all.sh
```

The serious $p = 4$ optimizer path can take hours and is cached in the current release artifacts.

10 Conclusion

This proof-of-concept establishes a clean finite-size baseline for a 19-site Kagome Heisenberg patch. A signed weighted-RVB initializer captures most of the exact-state structure, and a shallow edge-colored Heisenberg-HVA gives additional no-calibration circuit-compatible refinement. The best current no-calibration HVA result reaches

$$E = -29.037601351802, \quad F = 0.982518440088,$$

with an energy error of 0.10856676 relative to the exact benchmark. The result is strong enough to motivate deeper or more expressive Heisenberg-level ansatz work, but the present evidence remains finite-size and does not claim scalable hardware QSL preparation.

References

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